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The PNB - Nirav Modi Fraud: A Fraud Analytics Case Study

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1. Executive summary

From March 2011 to 2017, officials at Punjab National Bank's Brady House branch in Mumbai issued roughly 1,590 unauthorised bank guarantees to firms run by jeweller Nirav Modi and his uncle Mehul Choksi, leaving PNB liable for about ₹13,000–14,000 crore (~US$2 billion). It is widely described as the largest fraud in Indian banking history (Free Press Journal).

The fraud did not need hacking or forged paper. It used a gap between two of the bank's own systems: the SWIFT messaging platform, which sent guarantees to other banks, and the Core Banking System (CBS), which recorded them. Because the two were not linked, guarantees went out but never showed up in PNB's books.

We chose this case because it is rich in public data (CBI chargesheets, Parliament replies, PNB's results) and because every element of the assignment — the mechanism, the loophole, the detection and the aftermath — is clearly documented. It is also a textbook example for fraud analytics: a simple daily reconciliation of two datasets would have exposed it years earlier.

2. Background

The bank. Punjab National Bank (PNB) was, in 2018, India's second-largest state-owned lender. Brady House in South Mumbai was its mid-corporate branch and its second-biggest lending branch in the country (The Quint).

The accused. Nirav Modi founded Firestar Diamond in 1999 and later the Nirav Modi luxury jewellery brand, with stores in New York, London and Hong Kong. His uncle Mehul Choksi ran Gitanjali Gems, a large mass-market jewellery group. The Modi firms named as borrowers were Diamonds R US, Solar Exports and Stellar Diamonds; partners included Nirav Modi, his brother Nishal, his wife Ami and Choksi (CBI chargesheet, via Moneylife).

The inside men. Gokulnath Shetty, deputy manager in the foreign exchange department at Brady House, and Manoj Kharat, a single-window clerk who entered messages into SWIFT.

Key terms in plain words:

- **Letter of Undertaking (LoU):** a promise from one bank to another bank. It says, "Lend money to our customer; if they don't repay, we will." Importers used LoUs to get cheap short-term loans called *buyer's credit* from foreign branches of Indian banks, to pay overseas suppliers.
- **SWIFT:** the global messaging network banks use to send each other payment instructions and guarantees. A SWIFT message from PNB was trusted by any receiving bank as genuine.
- **Core Banking System (CBS):** PNB's internal ledger (Finacle software), where every loan, guarantee and limit is supposed to be recorded and monitored.

A normal LoU needs three things: a sanctioned credit limit, collateral or a cash margin, and an entry in CBS. The fraudulent LoUs had none of them.

3. What the fraud was about

In simple terms, PNB staff gave the Modi and Choksi firms bank guarantees worth thousands of crores without any approval, security or record — and foreign branches of other Indian banks lent money on the strength of those guarantees. When the loans were not repaid, PNB was legally on the hook.

On paper, the money was buyer's credit to import rough diamonds and pearls. In reality, investigators found that the funds went to "supplier" firms abroad that were set up and controlled by the accused themselves. Six Hong Kong companies alone received over ₹8,200 crore from the PNB-backed loans (The Quint).

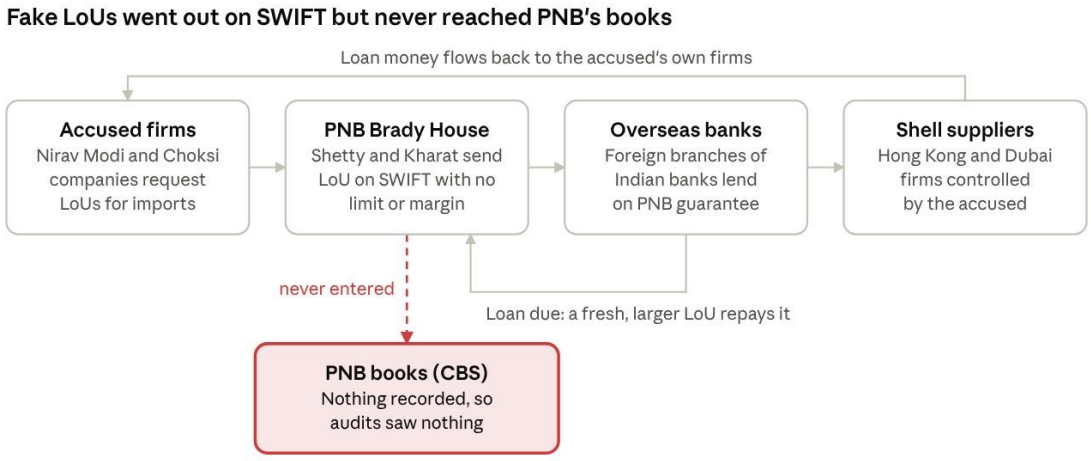
Metric	Value	Source
Period of fraud	March 2011 – 2017 (~7 years)	CBI, via Outlook
Fraudulent LoUs	1,213 to Nirav Modi group; 377 to Choksi / Gitanjali group	Deccan Herald
Amount — Nirav Modi firms	₹6,498.20 crore	CBI chargesheet, via Moneylife
Amount — Choksi firms	₹7,080 crore (PNB's claim)	Outlook
Fraud reported to RBI	₹12,967.86 crore (March 2018)	Business Standard
PNB's total liability to other banks	₹14,356 crore	Business Standard

The headline figure grew over time — ₹11,400 crore in February 2018, then about ₹12,700 crore, then over ₹14,000 crore — as PNB found more fraudulent messages. This growth is itself a lesson: the bank did not know its own exposure because the guarantees were never in its records.

4. How they did it — the modus operandi

The scheme ran like a revolving door: new fake guarantees paid off old loans, so nothing ever looked overdue.

1. **Request.** The Modi or Choksi firm asked the Brady House branch for an LoU to pay an overseas "supplier" for imported diamonds or pearls.
2. **Fake LoU sent on SWIFT.** Kharat typed the LoU message into SWIFT and Shetty authorised it — often using shared login credentials, so one person could effectively both create and approve. No credit limit was sanctioned and no cash margin or collateral was taken (CBI chargesheet, via Moneylife).
3. **Nothing recorded in CBS.** The transaction was deliberately not entered in the Core Banking System, so it never appeared in PNB's books, limit reports or audits.
4. **Overseas bank lends.** The foreign branch (for example, Allahabad Bank and Axis Bank in Hong Kong) saw a genuine SWIFT message from PNB and released buyer's credit into PNB's nostro account abroad, which was then paid to the "supplier" (The Quint).
5. **Money goes to the accused's own shell firms.** The "suppliers" in Hong Kong, Dubai and elsewhere were dummy companies controlled by the Modi group, so the money effectively came back to them.
6. **Roll over.** When a loan fell due, a fresh, larger LoU was issued to repay it plus interest. Liabilities snowballed, but no default ever appeared. LoUs were also issued for up to one year, longer than RBI rules allowed for such trade credit (DailyO).



LoU fraud cycle · 4 parties, 1 rollover loop

The rollover loop kept the fraud alive: each new guarantee paid off an old loan, and none of them ever reached PNB's books.

5. The loopholes they exploited

The core loophole was technical — SWIFT was not connected to CBS — but it only worked because five other controls also failed. RBI itself called the case operational risk from employee misconduct and a failure of internal controls (Business Standard / Reuters).

Loophole	What the rule said	What actually happened
SWIFT not integrated with CBS	Every SWIFT message should match a CBS entry	Staff had to log SWIFT activity manually; Shetty simply didn't, leaving no trail in the books
No daily reconciliation	Internal rules required SWIFT logs to be checked against CBS every day	The check was not done; PNB's own 162-page internal report said this alone would have caught the fraud (Onmanorama / Reuters)
Shared passwords, weak maker–checker	One person creates a message, a different person approves it	SWIFT credentials were shared, so the same people could send and approve guarantees
No staff rotation	PNB policy: no officer in one post for more than 3 years	Shetty stayed at Brady House for about 7 years until he retired in 2017
Audits and inspections missed it	Concurrent, internal, statutory and RBI audits should test trade-finance exposure	Since nothing was in CBS, auditors who checked the books saw nothing wrong
Overseas banks didn't verify	Lending banks should confirm limits and margins with the issuing bank	Branches abroad relied on the SWIFT message alone and kept lending for years

The top level also failed. The CBI alleged that PNB's senior management had prior warnings from RBI about SWIFT misuse but did not act on them (Business Today).

6. How they got caught

The fraud was not caught by any audit or analytics system — it was caught by accident, when the inside man was no longer there.

Gokulnath Shetty retired in May 2017. In mid-January 2018, Modi's firms Solar Exports, Stellar Diamonds and Diamonds R US came to the branch asking for fresh LoUs for buyer's credit, as usual (Business Today). The new officials asked for a 100% cash margin, which is standard practice. The firms' representatives objected that they had never been asked for one before.

That raised a simple question: where were the earlier LoUs? Staff searched the Core Banking System and found no record of any sanctioned limit or earlier guarantee. Around the same time, a couple of matured LoUs went unpaid and the overseas banks approached PNB for repayment (Business Today). An internal inquiry then traced the SWIFT messages and uncovered the full trail.

- **29 January 2018:** PNB filed a complaint with the CBI.
- **31 January 2018:** the CBI registered its FIR against Nirav Modi, Ami Modi, Nishal Modi, Mehul Choksi and others (Moneylife).
- **14 February 2018:** PNB disclosed the fraud to the stock exchanges (The Quint).

By then it was too late to stop the main accused. Nirav Modi and Mehul Choksi had both left India in the first week of January 2018, weeks before the complaint. Nirav Modi was even photographed at the World Economic Forum in Davos on 23 January.

7. What happened after

The fraud cost PNB its biggest-ever loss, changed RBI rules for every bank in India, and started an eightyear legal chase that is still not over.

Impact on PNB. PNB posted a net loss of ₹13,417 crore for January–March 2018, the biggest quarterly loss ever by an Indian lender, and set aside ₹7,178 crore — half of the ₹14,356 crore it owed other banks — in that quarter alone. For the full year FY2017-18 it lost ₹12,283 crore against a profit of ₹1,325 crore the year before (Business Standard). Its share price fell about 48% between January and mid-May 2018 (The Quint).

Changes in the system.

- RBI banned all banks from issuing LoUs and Letters of Comfort for import trade credit on 13 March 2018, with immediate effect (SCC Online).
- PNB added a third level of SWIFT re-authorisation at a central team in Mumbai, restricted SWIFT hours, and made concurrent auditors reconcile every SWIFT message with CBS daily (Business Standard).
- Banks across India were pushed to integrate SWIFT with CBS and enforce staff rotation.
- Parliament passed the Fugitive Economic Offenders Act, 2018, letting courts confiscate assets of offenders who flee India; Nirav Modi was declared a fugitive economic offender under it.

Legal action and current status (as of October 2026).

- **Bank staff and aides:** Shetty, Kharat and Modi's accountant Hemant Bhatt were among the first arrested (Hindustan Times). The CBI's chargesheet also named PNB's former MD and CEO Usha Ananthasubramanian, then heading Allahabad Bank, and other senior officials.

- **Nirav Modi:** arrested in London on 19 March 2019 and held in Wandsworth prison. A UK court ordered his extradition in February 2021. He has since lost every appeal; in March 2026 the High Court refused to reopen his case (News on AIR), and in July 2026 the European Court of Human Rights reportedly refused relief, leaving his surrender to India pending (Business Today).
- **Mehul Choksi:** took Antiguan citizenship before fleeing, and was arrested in Antwerp, Belgium, on 12 April 2025 (News on AIR). Belgian courts upheld his arrest, and in April 2026 the Antwerp Court of Appeal advised extradition on six of seven charges; the final decision rests with the Belgian government (Open).
- **Latest:** on 28 September 2026, Modi associate Sandeep Mistry was deported from the UAE and arrested by the CBI in Mumbai (Khaleej Times).

Date	Event
March 2011	First fraudulent LoU issued from Brady House
May 2017	Gokulnath Shetty retires
Early January 2018	Nirav Modi and Mehul Choksi leave India
Mid-January 2018	Fraud discovered when firms ask for new LoUs
29–31 January 2018	PNB complains to CBI; FIR registered
14 February 2018	PNB discloses fraud to stock exchanges
13 March 2018	RBI bans LoUs for import trade credit
15 May 2018	PNB reports ₹13,417 crore quarterly loss
19 March 2019	Nirav Modi arrested in London
25 February 2021	UK court orders Nirav Modi's extradition
12 April 2025	Mehul Choksi arrested in Antwerp
March–July 2026	Nirav Modi loses final UK and ECHR challenges
April 2026	Belgian appeals court advises Choksi's extradition
28 September 2026	Aide Sandeep Mistry deported from UAE to India

8. Fraud analytics lens: how data could have caught it earlier

Every signal needed to catch this fraud already existed in PNB's data; nobody joined the datasets together. The table below shows the checks a fraud analytics team would run and what each would have revealed.

Red flag / check	Data needed	Analytics technique	What it would have shown
SWIFT messages with no CBS entry	SWIFT outbound log + CBS guarantee register	Daily reconciliation (an "anti-join": SWIFT rows with no matching CBS row)	Every fraudulent LoU, from the first one in 2011
Guarantees rolled over to repay older ones	LoU amounts, dates, beneficiaries	Rule: new LoU issued near maturity of an old one for a similar or larger amount, same customer	A growing chain of LoUs repaying each other — a Ponzi-like pattern
Abnormal tenor	LoU validity periods	Distribution / outlier analysis	365-day LoUs where trade credit rules allowed much shorter terms
Customer concentration at one branch	LoU volume by customer and branch	Concentration ratio, trend over time	A handful of related jewellery firms dominating one branch's foreign exchange business
Related-party "suppliers"	Beneficiary names, addresses, directors	Network (graph) analysis, entity matching	Overseas suppliers linked to the borrowers themselves
Same user creating and approving	SWIFT user IDs, timestamps, terminals	User behaviour analytics, segregationof-duties checks	Shared credentials and one person effectively doing both steps

Staff in post too long	HR postings data	Exception report against the 3-year rotation rule	Shetty in the same sensitive role for ~7 years

Scale also mattered. PNB told Parliament it had issued 41,178 LoUs since 2011, of which 11,224 had a validity of 365 days and 2,204 ran beyond 365 days (Business Today). At that volume, manual checks are not enough; only automated, rule-based and anomaly-based monitoring can keep up.

The fraud triangle applied:

- **Pressure / motive:** the jewellery groups needed cheap, continuous funding to sustain rapid global expansion.

- **Opportunity:** SWIFT–CBS gap, shared passwords, no rotation, no reconciliation.
- **Rationalisation:** a long, cosy relationship with "prestigious" clients, and junior staff who saw themselves as just following orders.

9. Conclusion and key learnings

The PNB fraud was not a clever hack; it was an ordinary control gap left open for seven years. Two employees, one unconnected system and a skipped daily check were enough to move about ₹13,000 crore.

1. **Integrate systems, or reconcile them every day.** Any data that leaves the bank (SWIFT) must match the books (CBS). This single check would have stopped the fraud at the first LoU.
2. **Controls only work if they are enforced.** PNB had rules on reconciliation, password secrecy and staff rotation — they were simply not followed or monitored.
3. **Monitor patterns, not just transactions.** Each LoU looked normal on its own; the rollover chain, the tenors and the customer concentration were the real warning signs.
4. **Trust, but verify across institutions.** Lending banks accepted SWIFT messages at face value for years.
5. **Detection should not depend on luck.** This fraud surfaced only because the insider retired. A strong fraud analytics function finds such schemes through data, not chance.

For us as future banking and analytics professionals, the case shows that fraud analytics is less about complex models and more about asking a basic question of the data every day: *does what we sent out match what we recorded?*

References

1. Business Standard (PTI), "Fraud hit PNB posts biggest quarterly loss of Rs 13,417 crore in Q4", 15 May 2018. Link

2. Business Standard (PTI), "PNB takes steps to secure SWIFT operations", 13 March 2018. Link
3. Business Standard (Reuters), "Eyes wide shut — the $1.8 billion PNB fraud that went unnoticed", 19 February 2018. Link
4. Business Today, "Gem of a scam: How Nirav Modi used fake LoUs to defraud PNB", February 2018. Link
5. Business Today, "PNB fraud: Public sector bank issued over 41,000 LoUs since 2011", 23 March 2018. Link
6. Business Today, "CBI claims bank's top brass failed to prevent Nirav Modi scam", 15 May 2018. Link
7. Business Today, "Nirav Modi's extradition soon? European court rejects final legal challenge", 6 July 2026. Link
8. Moneylife (IANS), "PNB scam: CBI chargesheet names Allahabad Bank CEO Usha Ananthasubramanian, 21 others", May 2018. Link

9. Outlook India (PTI), "PNB scam: CBI files supplementary chargesheet against ex-official of Gitanjali Group", June 2021. Link
10. Deccan Herald, "PNB EDs connected with Nirav Modi", 2019. Link
11. Onmanorama (Reuters), "Lapses at many levels of bank led to huge PNB fraud, internal report shows", 20 June 2018. Link
12. SCC Online, "RBI discontinues Letter of Undertaking, Letter of Comfort as instruments of trade credit", 14 March 2018. Link
13. The Quint, "How a $1.8 Bn fraud at PNB was detected", February 2018. Link
14. News on AIR, "Nirav Modi faces setback as UK High Court rejects plea to reopen extradition case", 26 March 2026. Link
15. Open Magazine, "Mehul Choksi's escape nears end? Belgium court clears path for India return", 12 April 2026. Link
16. Khaleej Times, "Fugitive in Nirav Modi bank fraud case deported from UAE", September 2026. Link